

An Exactly Conservative Semi-Lagrangian Scheme (CIP-CSL) in One Dimension

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ABSTRACT

Two semi-Lagrangian schemes that guarantee exactly mass conservation are proposed. Although they are in a nonconservative form, the interpolation functions are constructed under the constraint of conservation of cell-integrated value (mass) that is advanced by remapping the Lagrangian volume. Consequently, the resulting schemes conserve the mass for each computational grid cell. One of them (CIP-CSL4) is the direct extension of the original cubic-interpolated propagation (CIP) method in which a cubic polynomial is used as the interpolation function and the gradient is calculated according to the differentiated advection equation. A fourth-order polynomial is employed as the interpolation function in the CIP-CSL4 method and mass conservation is incorporated as an additional constraint on the reconstruction of the interpolation profile. In another scheme (CIP-CSL2), the CIP principle is applied to integrated mass and the interpolation function becomes quadratic. The latter one can be readily extended to multidimensions. Besides the linear advection-transportation equation, these schemes are also applied to the nonlinear advection problem with a large Courant–Freidrichs–Lewy number.

1. Introduction

Solving atmospheric problems together with oceanic ones and/or solid structure is a grand challenge to the field of computational mathematics. For these types of problems, such as snow melting and deformation, and water evaporation, we need to treat topology and phase changes of the materials simultaneously, where the grid system aligned to the solid or liquid surface has no meaning and sometimes the mesh is distorted and even broken up. A universal treatment of all phases by one simple algorithm is thus essential and we are at the turning point of attacking this goal.

Even without considering phase change, problems of surface capturing and structure–fluid interaction are not easy tasks. In most cases, the grid cannot always be adapted to those surfaces. Therefore, the description of moving surfaces of complicated shapes in the Cartesian grid system will be a challenging subject.

In order to attack the problems mentioned above, we must first find a method to treat a sharp interface and to solve the interaction of compressible gas with an incompressible liquid or solid. Toward this goal, we take

an Eulerian approach based on the cubic-interpolated propagation (CIP) method (Takewaki et al. 1985; Takewaki and Yabe 1987; Yabe and Takei 1988; Yabe and Aoki 1991; Yabe et al. 1991), which does not need an adaptive grid system and therefore removes the problems of grid distortion caused by structural breakup and topology change. The material surface can be captured by almost one grid throughout the computation (Yabe and Xiao 1993; Yabe and Xiao 1995; Xiao et al. 1997). Furthermore, the scheme can treat all the phases of matter from solid state through liquid and two phase states to gas without restriction on the time step from high sound speed (Yabe and Wang 1991). A pressure-based algorithm coupled with a semi-Lagrangian approach like the CIP proved to be stable and robust in analyzing these subjects.

As an efficient and accurate approach to computing the advection process, semi-Lagrangian schemes have been extensively studied and widely incorporated into numerical models for atmospheric flows during the past decades. Surveys of such a topic were given by Staniforth and Côté (1991) and Smolarkiewicz and Pudykiewicz (1992).

Unlike most Eulerian methods, which are derived by using control volume or cell-integrated average, the Lagrangian invariant solution used in a semi-Lagrangian scheme is based only on spatial points and is always cast in a nonconservative form. Few studies on the con-

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servation of the semi-Lagrangian schemes have been reported so far in the literature because of the intrinsic difficulties in introducing a conservative formulation in a semi-Lagrangian scheme. A quasi-conservative semi-Lagrangian advection scheme is proposed by Priestley (1993) where a semi-Lagrangian scheme is derived by weighting the flux-corrected transport method (Boris and Book 1973; Zalesak 1979) and a higher-order interpolation (Bermejo and Staniforth 1992) through a posteriori restoration. Another direct but practical approach to constructing a conservative semi-Lagrangian scheme is to extend some existing Eulerian methods to a formulation that can be stable even under a large Courant number. Lin and Rood (1996) suggested the so-called flux-form semi-Lagrangian schemes using the piecewise parabolic method (PPM) (Colella and Woodward 1984) and the monotone upstream-centered scheme for conservation laws (MUSCL) (van Leer 1974, 1977) methods. For an arbitrary time step, the numerical flux is decomposed as an integer part and a fraction part, and a relatively simple expression for computing the flux on cell interfaces over a long time step can be obtained by using an averaged velocity along trajectory. As the conservation of the advected physical field is important or appears to be essential in many applications, development of a semi-Lagrangian scheme that guarantees this property should be worthy of further efforts.

Our recent work leads to some new improvements to the conservative property of the CIP method. We find that an exactly conservative semi-Lagrangian (CSL) scheme can be derived by constraining the reconstruction of the interpolation. A recent version of the CIP-CSL4 (Tanaka et al. 2000) can provide an exactly conservative semi-Lagrangian scheme. Since the scheme does not use the cubic polynomial but uses different orders of polynomials, we redefine the name of these CIP families as constrained interpolation profiles and still keep the abbreviation CIP. This means that various constraints such as the time evolution of spatial gradient, which is used in the original CIP method, or spatially integrated conservative quantities can be used to construct the profile. Based on the CIP concept, another variant, the CIP-CSL2 scheme, is proposed using a quadratic interpolation function. In this paper, we shall give a brief review of the CIP method and then propose a method to incorporate the conservative property in semi-Lagrangian schemes.

2. Review of the CIP method

a. Numerical difficulty

When implementing a numerical simulation, a discretization process is generally unavoidable. Such a process always causes the loss of the “subgrid” information due to the finite truncation of grid resolution. The primary goal of a numerical algorithm will be to retrieve

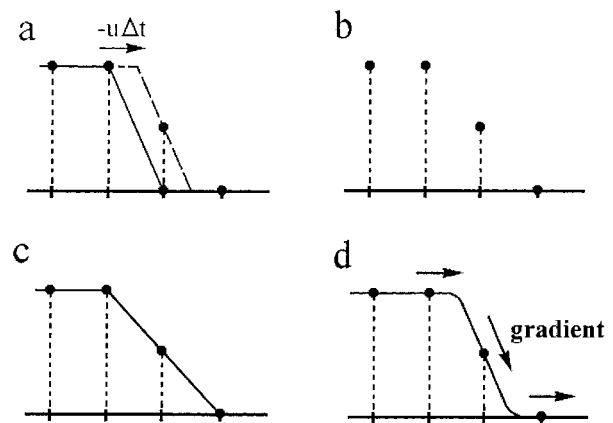


FIG. 1. The principles of the CIP method are as follows. (a) The solid line is an initial profile and the dashed line is an exact solution after advection, whose solution (b) is at discretized points. (c) When (b) is linearly interpolated, numerical diffusion appears. (d) In the CIP, the spatial derivative also propagates and the profile inside a grid cell is retrieved.

the lost information inside the grid cell between these scattered points. Most numerical schemes proposed before, however, did not take care of the real solution inside the grid cell and the resolution has been limited to the grid size. The CIP method proposed by one of the authors tries to construct a solution inside the grid cell close enough to this real solution of the given equation with some constraints. We explain here the strategy behind this method by using an advection equation:

$$\frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} = 0. \quad (1)$$

When the velocity is constant, the solution of Eq. (1) gives a simple translational motion of a wave with a velocity u . The initial profile (solid line in Fig. 1a) moves like the dashed line in a continuous representation. At this time, the solution at grid points is denoted by circles and is the same as the exact solution. However, if we eliminate the dashed line as in Fig. 1b, then the information from the profile inside the grid cell is lost and it is hard to imagine the original profile, and it is natural to imagine a profile like that shown by solid line in Fig. 1c. Thus, numerical diffusion arises when we construct the profile by the linear interpolation even with the exact solution as shown in Fig. 1c. This process is called the first-order upwind scheme (Courant et al. 1952). On the other hand, if we use quadratic polynomial for interpolation, it suffers from overshooting. This process is the Lax–Wendroff scheme (Lax and Wendroff 1960).

What made this solution worse? It is because we neglected the behavior of the solution inside a grid cell and merely followed after the smoothness of the solution. From this experience, we understand that a method incorporating the real solution into the profile within a grid cell is quite an important subject. We propose to

approximate the profile as shown below. Let us differentiate Eq. (1) with spatial variable x , then we get

$$\frac{\partial g}{\partial t} + u \frac{\partial g}{\partial x} = -\frac{\partial u}{\partial x} g, \quad (2)$$

where g stands for the spatial derivative of f , $\partial f / \partial x$. In the simplest case where the velocity u is constant, Eq. (2) coincides with Eq. (1) and represents the propagation of a spatial derivative with a velocity u . By this equation, we can trace the time evolution of f and g on the basis of Eq. (1). If g could be predicted to propagate as is shown by the arrows in Fig. 1d, the profile after one step would be limited to a specific profile. It is easy to imagine that, by this constraint, the solution becomes much closer to the initial profile that is the real solution. Most importantly, the solution thus created gives a profile consistent with Eq. (1) even inside the grid cell. The importance of this consistency will be demonstrated in the next section.

If both the values of f and g are given at two grid points, the profile between these points can be interpolated by cubic polynomial $F(x) = ax^3 + bx^2 + cx + d$. Thus, recalling that a constant advecting velocity has been assumed, the profile at the $n + 1$ step is readily obtained by shifting the profile by $u\Delta t$ like $f^{n+1} = F(x - u\Delta t)$, $g^{n+1} = dF(x - u\Delta t)/dx$:

$$\begin{aligned} a_i &= \frac{g_i + g_{iup}}{\Delta x_i^2} + \frac{2(f_i - f_{iup})}{\Delta x_i^3}, \\ b_i &= \frac{3(f_{iup} - f_i)}{\Delta x_i^2} - \frac{2g_i + g_{iup}}{\Delta x_i}, \\ \Delta x_i &= x_{iup} - x_i \quad iup = i - \text{sgn}(u_i) \end{aligned} \quad (3)$$

$$f_i^{n+1} = a_i \xi_i^3 + b_i \xi_i^2 + g_i^n \xi_i + f_i^n,$$

$$g_i^{n+1} = 3a_i \xi_i^2 + 2b_i \xi_i + g_i^n, \quad (4)$$

where we define $\xi_i = -u_i \Delta t$ and $\text{sgn}(u)$ stands for the sign of u . We constructed (3) over a neighboring cell of the interested grid i , which implies $u_i \Delta t / \Delta x \leq 1$, but in no way limit ourselves to this specific case. As can be seen later, the CIP scheme allows the use of a longer time step like other semi-Lagrangian methods. Figure 2a shows a profile after 1000 steps with this CIP method for the propagation of a square wave. In this example, Courant (Courant–Freidrichs–Lewy (CFL) number is 0.1, and the initial values of g_i^0 on all the grid points are estimated using a central difference formulation from the given f_i^0 .

Although there exist various polynomial functions such as linear, quadratic Lagrange, cubic Lagrange, cubic spline, and quintic Lagrange (Staniforth and Côté 1991), all of these schemes (except those using the linear interpolation function) need at least three points in constructing interpolation approximations in one dimension. A more compact scheme by which one can construct an interpolation function of high accuracy with

less computational stencils is desired in many situations, such as the calculations of discontinuities or large gradients. Moreover, in a model with a limited computational domain, different approximations for the derivatives have to be used at the grid points close to the boundaries; these approximations are usually of lower order than the approximations used deeper in the interior. Thus, a scheme that uses less stencils may be advantageous in treating computational boundaries since less boundary points need to be handled. Another attractive feature of reducing stencils may be the reduction in data transfer in parallel implementations on distributed memory architectures. In this sense, the CIP seems to be attractive since it uses only one cell for computation even in three dimensions.

The CIP is completely different from conventional semi-Lagrangian methods regarding the computation of the derivatives. In the latter methods, as mentioned above, the gradient is calculated based on the function values at neighboring grid points by either assuming the continuity of the quantity, the first- and sometimes the second-order derivatives of the quantity at the mesh boundaries (Purnell 1976), or using approximations based on local grid points (Williamson and Rasch 1989). By special treatment of the first derivatives of the interpolation function, the CIP achieves a compact form that uses only one mesh cell to construct the interpolation profile and provides subcell resolution.

b. Phase error of CIP

From the numerical experiments, we always observe that the CIP method gives very good phase speed with very small numerical dispersion even when applied to the discontinuous profile where high wavenumber components are dominant. It is interesting to examine the phase error of various schemes using the method proposed by Purnell (1975) and Utsumi et al. (1996). Figure 3 summarizes those results. As is well known, the phase speed of conventional schemes departs from the exact one, which is shown by the solid line, around $k\Delta x = \pi/2$. Surprisingly, however, the CIP can reproduce the correct phase speed even up to $k\Delta x = \pi$. This is remarkable because $k\Delta x = \pi$ means that one wavelength is described by three grid points. Let us consider the case shown in Fig. 4, where values of the three points are zero. Even in this case, one wave can exist as shown in the figure. The CIP gives correct spatial gradients, which are nonzero at these points, and therefore it recognizes the existence of the wave inside the grid cell. Any code that uses only the information of the value, which is zero now, cannot correctly recognize the wave even if a higher-order polynomial is employed.

The importance of Eq. (2), which predicts the propagation of gradients, can be clearly demonstrated in comparison with the cubic spline (Purnell 1975) and the PPM (Colella and Woodward 1984) methods, both of which have third-order accuracy. Although the cubic

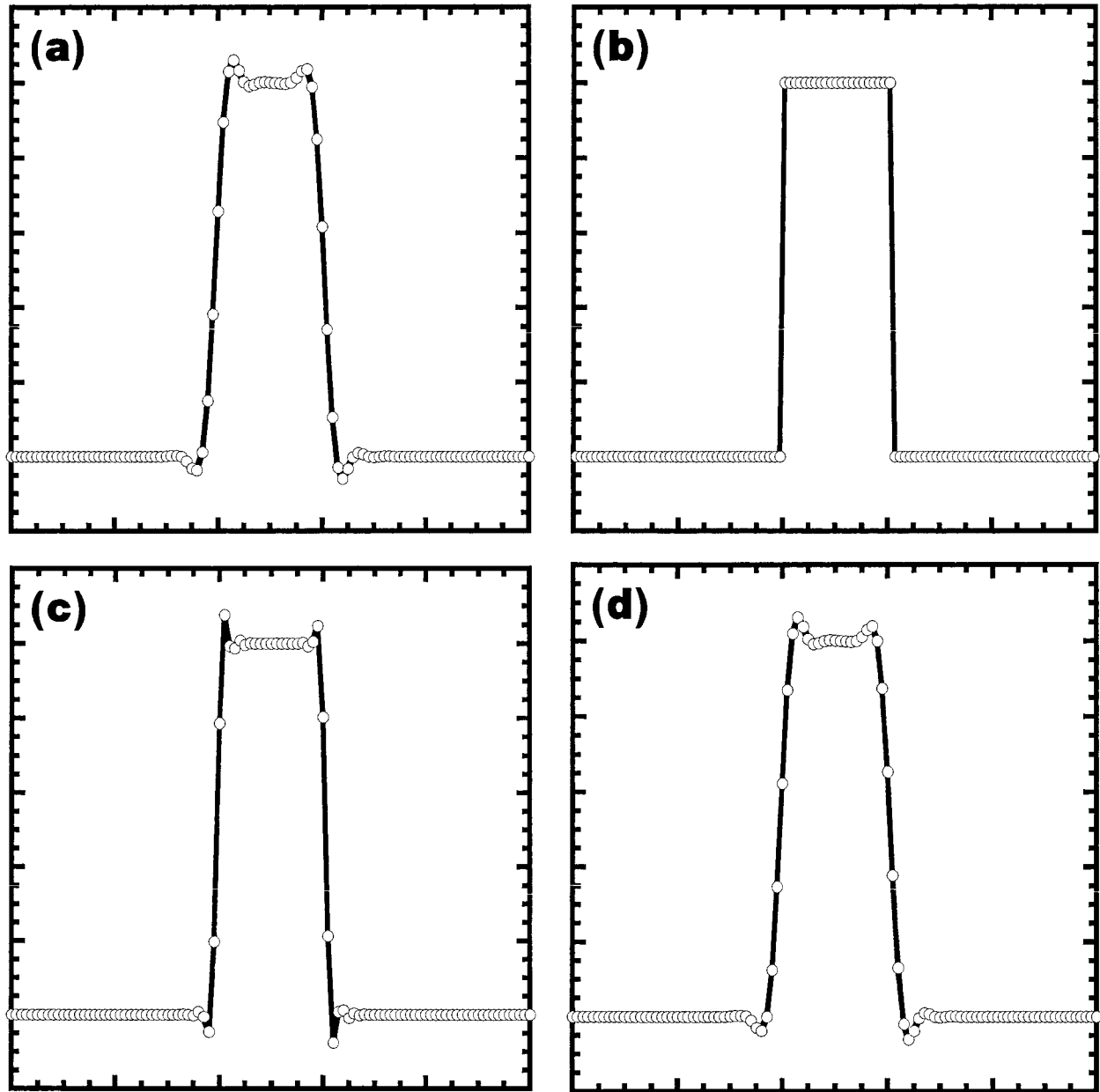


FIG. 2. Square wave propagation with (a) CIP, (b) CIP with a tangent transformation, (c) CIP-CSL4, and (d) CIP-CSL2.

spline uses the same cubic polynomial as the CIP, it cannot reproduce the results of the CIP, because the gradient of the spline is determined merely from the smoothness requirement. As is easily recognized, such a constraint that is independent of the original equation will not help to retrieve the profile inside the grid cell.

There may be some objection that the CIP uses both a function and its derivative and that it seems to be the same as other schemes that use twice as many mesh points. We should address the following points against this objection.

1) The cubic spline shown in Fig. 3 uses both function

and its derivative like the CIP but fails to give a similar result.

2) The calculation costs do not increase even if an additional variable like the derivative is introduced. For example, the CPU time required for a simple advection problem in one dimension is spline/CIP = 1.68 (the Thomas method is used to solve the matrix), RCIP/CIP = 1.77, and PPM/CIP = 2.31. In multiple dimensions, spline becomes slower due to the matrix solution; for example, spline/CIP = 2.0 in two dimensions. Here RCIP means the CIP with the rational function that makes CIP monotone (Xiao et al. 1996).

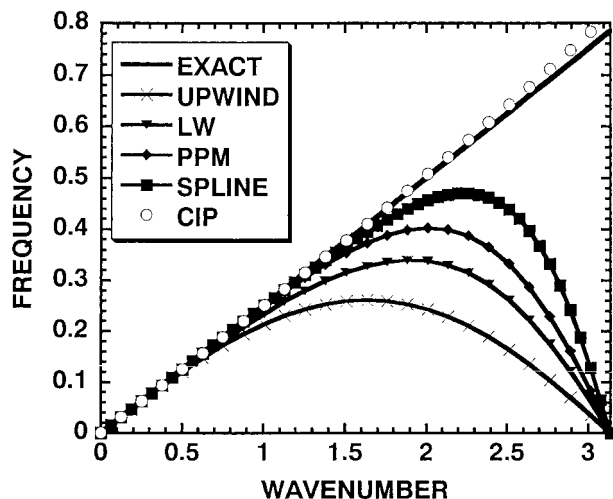


FIG. 3. Phase error of various schemes such as first-order upwind, Lax–Wendroff, PPM, spline, and CIP.

- 3) The memory increase owing to additional variables is less significant for higher dimensions. Let L_t be the number of time-dependent variables, then memory increase as $(\alpha + 1)L_t$ in the CIP and as $n^{\alpha+1}L_t$ in conventional schemes, where α is the dimension and n is the mesh refinement. Those will be $4L_t$ and $16L_t$ in three dimensions for the CIP and two times refined ($n = 2$) conventional schemes, respectively.
- 4) Adding to point 3, in most practical applications, we use additional variables like thermal conductivity, viscosity, temporal variables for matrix solution, and so on. Let L_0 be the number of these variables; then the memory requirement is $(\alpha + 1)L_t + L_0$ and $n^{\alpha+1}L_t + L_0$. Since $L_0 \gg L_t$, usually the memory requirement of the CIP is similar to other schemes even for unrefined mesh $n = 1$.

The accurate prediction of the phase velocity enables the CIP method to be useful in various applications. For example, the CIP can be used as an efficient scheme for capturing a sharp interface. Yabe and Xiao (1993, 1995) devised an interface tracking technique that appears efficient, geometrically faithful, and diffusionless. This interface tracking method is a combination of the CIP advection solver and a tangent function transformation. The method does not involve any interface construction procedure and is quite economical in computational complexity, especially for 3D applications. Figure 2b shows a 1D square wave propagation computed by the CIP method together with a tangent transformation. The initial sharpness is well preserved and the discontinuities are advected with the correct speed.

3. Toward a conservative semi-Lagrangian scheme

The CIP method treats the advection term separately from other terms in general hyperbolic equations. This enables us to use a large CFL number ($\equiv u\Delta t/\Delta x$) by

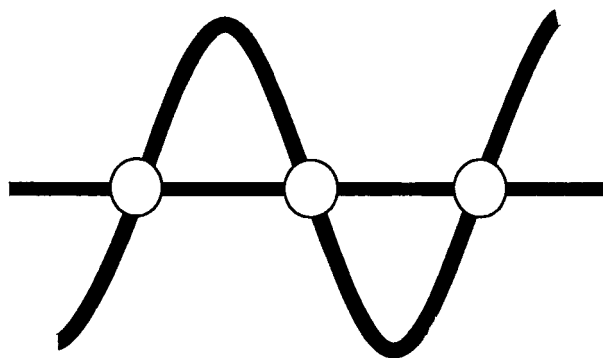


FIG. 4. The CIP can correctly recognize one wavelength with three grids points.

simply applying the procedure of Eq. (4) to the far-upstream grid cell from which the Lagrangian particle started to the present position of concern. Thus Eq. (4) is modified to be

$$\begin{aligned} f_i^{n+1} &= a_m \langle \xi \rangle^3 + b_m \langle \xi \rangle^2 + g_m \langle \xi \rangle + f_m^n, \\ g_i^{n+1} &= 3a_m \langle \xi \rangle^2 + 2b_m \langle \xi \rangle + g_m^n, \end{aligned} \quad (5)$$

where m is the grid point determined by

$$\begin{aligned} x_m &< x_p < x_{m+1}, & \text{for } u \leq 0 \\ x_{m-1} &< x_p < x_m, & \text{for } u > 0, \end{aligned}$$

and x_p is the particle position of upstream departure point calculated by

$$x_p = x_i + \int_{t+\Delta t}^t u \, dt. \quad (6)$$

This time integration is performed along the particle trajectory. Thus, $\langle \xi \rangle$ is the distance between these two points:

$$\langle \xi \rangle = x_p - x_m. \quad (7)$$

It should be noticed that ξ is neither $-u\Delta t$ nor $\int_{t+\Delta t}^t u \, dt$. In Eq. (5), a_m , b_m are given by simply replacing i by m in Eq. (3). We then get a semi-Lagrangian scheme that permits a large time step free from the restriction $\Delta t \leq u/\Delta x$.

Even in multiple dimensions, the strategy to find this upstream departure point is common to all other semi-Lagrangian schemes. Although the semi-Lagrangian algorithms can significantly reduce the computation time particularly in parallel computation, the loss of exact mass conservation, stemming from the fact that semi-Lagrangian schemes are generally not in a conservation form, inhibits them from long-term atmospheric and oceanic applications. In this section, we shall propose a method to recover this mass conservation in such semi-Lagrangian schemes.

We shall present two conservative schemes, namely CIP–CSL4 and CIP–CSL2. The CIP–CSL4 method was discussed in our previous paper (Tanaka et al. 2000).

We include it in the present paper because it provides a good comparison to the CIP-CSL2 in both scheme construction and numerical experiments.

a. CIP-CSL4 (Tanaka et al. 2000)

We discuss at first how to compute the conservation law such as

$$\frac{\partial f}{\partial t} + \frac{\partial(uf)}{\partial x} = 0. \quad (8)$$

As already seen, the CIP adopted an additional constraint, which is the spatial gradient, to represent the profile inside the grid cell. In order to endow the CIP method with the conservative property, we here added another constraint on the integrated value of f as

$$\int_{x_{i-1}}^{x_i} f^n dx = \rho_{i-1/2}^n. \quad (9)$$

Thus, a spatial profile can be constructed to satisfy this additional constraint as well. It shows that although f is advanced in a nonconservative form, conservation can be enhanced by constraining the cell-integrated quantity of the reconstructed interpolation profile. If we evaluate the cell-integrated quantity ρ by remapping the Lagrangian

volume as is shown later, we will get a numerical scheme that automatically satisfies the conservation of the cell-integrated quantity ρ .

Keeping this point in mind, then the i th function piece $F_i(x)$ must be determined so as to satisfy the following constraints:

$$\begin{aligned} F_i(x_{i-1}) &= f(x_{i-1}), & F_i(x_i) &= f(x_i) \\ \partial F_i(x_{i-1})/\partial x &= g(x_{i-1}), & \partial F_i(x_i)/\partial x &= g(x_i) \\ \int_{x_{i-1}}^{x_i} F_i(x) dx &= \rho_{i-1/2}. \end{aligned} \quad (10)$$

In order to meet the above constraints, a fourth-order polynomial can be chosen as the interpolation function $F_i(x)$. After $F_i(x)$ is determined, the time development of f and g is calculated simply by shifting the interpolation function $F_i(x)$ by $u\Delta t$ in the same way as Eq. (4) of the CIP method as follows:

$$\begin{aligned} f_i^* &= F_i(x_i - u_i^n \Delta t) \\ &= a_i^n \xi_i^4 + b_i^n \xi_i^3 + c_i^n \xi_i^2 + g_i^n \xi_i + f_i^n, \end{aligned} \quad (11)$$

$$\begin{aligned} g_i^* &= \partial F_i(x_i - u_i^n \Delta t)/\partial x \\ &= 4a_i^n \xi_i^3 + 3b_i^n \xi_i^2 + 2c_i^n \xi_i + g_i^n, \end{aligned} \quad (12)$$

where $\xi_i = -u_i^n \Delta t$, and

$$\begin{aligned} a_i^n &= \frac{-5[6(f_{iup} + f_i)\Delta x_i - (g_{iup} - g_i)\Delta x_i^2 + 12\text{sgn}(u_i^n)\rho_{i-\text{sgn}(u_i^n)/2}]}{2\Delta x_i^5}, \\ b_i^n &= \frac{4[(7f_{iup} + 8f_i)\Delta x_i - (g_{iup} - (3/2)g_i)\Delta x_i^2 + 15\text{sgn}(u_i^n)\rho_{i-\text{sgn}(u_i^n)/2}]}{\Delta x_i^4}, \\ c_i^n &= \frac{-3[4(2f_{iup} + 3f_i)\Delta x_i - (g_{iup} - 3g_i)\Delta x_i^2 + 20\text{sgn}(u_i^n)\rho_{i-\text{sgn}(u_i^n)/2}]}{2\Delta x_i^3}, \\ \Delta x_i &= x_{iup} - x_i, \quad iup = i - \text{sgn}(u_i^n). \end{aligned} \quad (13)$$

The above procedure can be easily extended to a high CFL number, $u\Delta t/\Delta x > 1$, such as Eq. (5). Including such a case, the time evolution of ρ is determined from the volume formed by four upstream departure points (in two dimensions) as shown in Fig. 5, where the region B is remapped into the region A. This mapping is complete, and thus each volume of upstream points does not overlap. In one dimension, this remapping is simply described by

$$\begin{aligned} \rho_{i-1/2}^{n+1} &= \int_{x_{p(i-1)}}^{x_{m(i-1)}} f dx + \cdots + \int_{x_{m(i)-1}}^{x_{p(i)}} f dx \\ &= \Delta\rho_{m(i-1)} + \cdots + [\rho_{m(i)-1/2} - \Delta\rho_{m(i)}], \end{aligned} \quad (14)$$

where $x_{p(i)}$ is the upstream departure point of x_i and is defined by

$$x_{p(i)} = x_i + \int_{t+\Delta t}^t u dt \quad (15)$$

and $x_{m(i)}$ is the point defined by

$$\begin{aligned} x_{m(i)} &< x_{p(i)} < x_{m(i)+1}, & \text{for } u \leq 0 \\ x_{m(i)-1} &< x_{p(i)} < x_{m(i)}, & \text{for } u > 0. \end{aligned}$$

In the last expression of Eq. (14), we defined

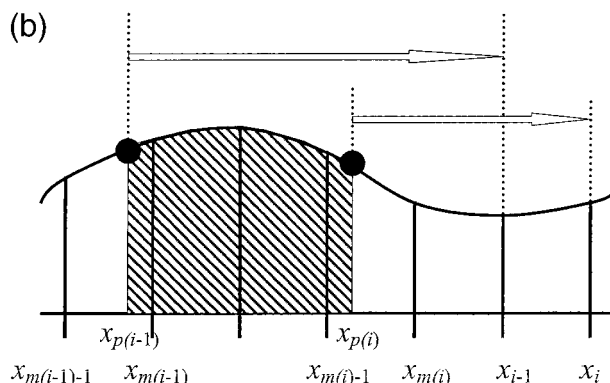
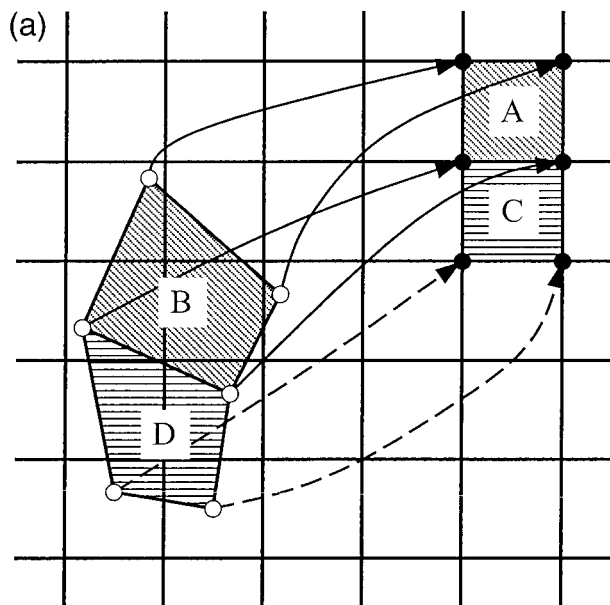


FIG. 5. (a) Remapping procedure gives the time evolution of ρ . The region B can be found by tracing the trajectories of vertices. (b) Remapping procedure in one dimension.

$$\begin{aligned} \Delta \rho_{m(i)}^n &= \int_{x_{p(i)}}^{x_{m(i)}} F_{m(i)}(x) dx \\ &= - \left[\frac{a_{m(i)}^n}{5} \langle \xi \rangle^4 + \frac{b_{m(i)}^n}{4} \langle \xi \rangle^3 + \frac{c_{m(i)}^n}{3} \langle \xi \rangle^2 \right. \\ &\quad \left. + \frac{g_{m(i)}^n}{2} \langle \xi \rangle + f_{m(i)}^n \langle \xi \rangle \right], \end{aligned} \quad (16)$$

where each coefficient is equivalent to Eq. (13) at the time step n and $\langle \xi \rangle$ is defined by

$$\langle \xi \rangle = x_{p(i)} - x_{m(i)}. \quad (17)$$

In the simplest case of a low CFL number, $u\Delta t/\Delta x < 1$, Eq. (14) is written as

$$\rho_{i-1/2}^{n+1} = \Delta \rho_{i-1}^n + (\rho_{i-1/2}^n - \Delta \rho_i^n). \quad (18)$$

The result of the square wave propagation is shown in

Fig. 2c. The initial value for the cell-integrated mass $\rho_{i-1/2}^0$ is calculated by a simple trapezoidal integration of the initial grid point values of f^0 as

$$\rho_{i-1/2}^0 = \frac{1}{2}(f_{i-1}^0 + f_i^0)\Delta x. \quad (19)$$

b. CIP-CSL2

The previous section gave a method to incorporate the mass conservation into a semi-Lagrangian scheme by simply extending the cubic interpolation in the CIP to a fourth-order polynomial. Although the increase of the order of the polynomial improved the accuracy of the solution, the extension to multidimensions may become expensive in computational cost. We here propose an alternative method for keeping the polynomial cubic. We shall use the same system of equation as Eqs. (8) and (9).

In the CIP, the time evolutions of f and $g = \partial f/\partial x$ are used as constraints to define a cubic polynomial, while in the CIP-CSL4, constraints are now f , $\partial f/\partial x$, and $\int f dx$ giving a fourth-order polynomial. It would be interesting to find a way to apply the CIP to the integrated value of f instead of f itself. The motivation to employ this analogy stems from the following advection equation:

$$\frac{\partial D}{\partial t} + u \frac{\partial D}{\partial x} = 0. \quad (20)$$

Interestingly, if we take a spatial derivative of Eq. (20) and define $D' \equiv \partial D/\partial x$, we obtain a conservative-type equation:

$$\frac{\partial D'}{\partial t} + \frac{\partial(uD')}{\partial x} = 0. \quad (21)$$

Since Eq. (21) is the same as Eq. (8), then we come to the idea of using $D' = f$ in Eq. (21) and $D = \int f dx$ in Eq. (20). This procedure is exactly the same as Eq. (1) by simply replacing f by $\int f dx$, together with Eq. (2) in which g is replaced by f . Thus all of the CIP procedure can be used for a pair of $\int f dx$ and f instead of f and $\partial f/\partial x$.

By this analogy, we shall introduce a function

$$D_i(x) = \int_{x_i}^x f(x') dx'. \quad (22)$$

We shall use a cubic polynomial to approximate this profile:

$$D_i(x) = A1_i X^3 + A2_i X^2 + f_i^n X, \quad (23)$$

where $X = x - x_i$. The role of the spatial gradient g in the CIP method is now played by f , which is the spatial gradient of $D(x)$ in the present scheme. By using the above relation, a profile of $f(x)$ between x_i and x_{iup} is then given by taking the derivative of Eq. (23):

$$f(x) = \frac{\partial D_i(x)}{\partial x} = 3A1_i X^2 + 2A2_i X + f_i^n. \quad (24)$$

From the definition of D in Eq. (22), it is clear that

$$D_i(x_i) = 0, \quad D_i(x_{iup}) = -\text{sgn}(u_i)\rho_{icell}^n, \quad (25)$$

where ρ_{icell}^n is the total mass of upwind cell defined at the cell center $i \pm 1/2$ and $icell = i - \text{sgn}(u_i)/2$. Since $\partial D/\partial x$ gives a value f , it is also clear that

$$\frac{\partial D_i(x_i)}{\partial x} = f_i^n, \quad \frac{\partial D_i(x_{iup})}{\partial x} = f_{iup}^n. \quad (26)$$

Therefore, the coefficients $A1_i$ and $A2_i$ are determined so as to satisfy the constraints of Eqs. (25) and (26). As a result of above simultaneous equations, the coefficients are determined explicitly as follows:

$$A1_i = \frac{f_i^n + f_{iup}^n}{\Delta x_i^2} + \frac{2\text{sgn}(u_i)\rho_{icell}^n}{\Delta x_i^3},$$

$$A2_i = -\frac{2f_i^n + f_{iup}^n}{\Delta x_i} - \frac{3\text{sgn}(u_i)\rho_{icell}^n}{\Delta x_i^2}, \quad (27)$$

where $\Delta x_i \equiv x_{iup} - x_i$. The above procedure applies also to a large CFL by simply considering the cell in Eq. (22) to be the cell at the upstream departure point. The calculation of the cell density ρ is performed in a similar way to that in section 3a. In the following, we shall give an explicit form of $\Delta\rho_i$ for a low CFL case,

$$\Delta\rho_i = \int_{x_i+\xi}^{x_i} f(x') dx' = -D_i(x_i + \xi)$$

$$= -(A1_i \xi^3 + A2_i \xi^2 + f_i^n \xi), \quad (28)$$

then the time development of ρ can be calculated from Eq. (18) with the aid of Eq. (28). It is interesting to observe by comparing Eq. (27) with Eq. (3) that the role of f, g in Eq. (3) is played by $D, \dot{f}$, respectively, because $D_i(x_i) - D_i(x_{iup}) = \text{sgn}(u_i)\rho_{icell}^n$ from Eq. (25). As expressed in Eq. (5), the solution procedure discussed here also applies to a large CFL number only if we construct D over the single grid cell covering the departure point as discussed at the beginning of this section.

Then, let us turn to the time evolution of the physical value f . We calculate the physical value f in the same way as the original CIP scheme. The conservative equation is rewritten as

$$\frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} = G, \quad (29)$$

where $G \equiv -f\partial u/\partial x$. Equation (29) is a simple advection equation for f . On the basis of the time-splitting algorithm of the CIP scheme (Yabe and Takei 1988; Yabe and Aoki 1991), we split the solution of Eq. (29) into two phases:

$$\text{advection phase,} \quad \frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} = 0; \quad (30)$$

$$\text{nonadvection phase,} \quad \frac{\partial f}{\partial t} = G. \quad (31)$$

After the advection phase is solved, the nonadvection phase is calculated with the result of the advection phase.

In the advection phase, we make use of the local analytic solution of Eq. (30), which is well known as the Lagrangian invariant solution $f(x_i, t + \Delta t) = f(x_i - u_i \Delta t, t)$. Since the profile of $f(x)$ between x_i and x_{iup} is given by Eq. (24), the solution of the advection phase f^* is calculated as

$$f_i^* = \frac{\partial D_i(x_i + \xi)}{\partial x} = 3A1_i \xi^2 + 2A2_i \xi + f_i^n. \quad (32)$$

After the advection phase is calculated by using Eq. (32) above, the result of the advection phase f^* is advanced to the value of the next time step f^{n+1} in the nonadvection phase. The nonadvection phase can be solved by using the conventional finite-difference method as follows:

$$f_i^{n+1} = f_i^* + G \Delta t, \quad (33)$$

where $G = -f_i^* (\partial u / \partial x)_i$ and the spatial derivative of the velocity $\partial u / \partial x$ is approximated by the simple centered finite difference. Figure 2d shows the result of the square wave propagation after 1000 time steps. We see from Fig. 2 that the CIP-CSL2 scheme provides quite a similar result to that of the original CIP.

c. Generalization

It would be useful to summarize all of this procedure here although the procedure is available in other references and was already given in the previous sections. The basic equations are now

$$\frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} = -f \frac{\partial u}{\partial x} + S \equiv G, \quad (34)$$

$$\frac{\partial f'}{\partial t} + u \frac{\partial f'}{\partial x} = -u' f' + G', \quad (35)$$

where G includes part of mass flux $\partial(u f) / \partial x$ in Eq. (8) and/or various source terms like diffusion. These equations are split into two phases:

1) advection phase,

$$\frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} = 0, \quad \frac{\partial f'}{\partial t} + u \frac{\partial f'}{\partial x} = 0; \quad (36)$$

2) nonadvection phase,

$$\frac{\partial f}{\partial t} = G, \quad \frac{\partial f'}{\partial t} = -u' f' + G'. \quad (37)$$

The advection phase is solved by, for CIP-CSL4,

TABLE 1. Grid refinement analysis of the schemes.

Scheme	$E_{\text{TOT}} \Delta x = 4.0$	$E_{\text{TOT}} \Delta x = 2.0$	$E_{\text{TOT}} \Delta x = 1.0$
CIP	1.570×10^{-2}	9.765×10^{-3}	6.041×10^{-3}
CIP-CSL2	1.562×10^{-2}	9.715×10^{-3}	6.012×10^{-3}
CIP-CSL4	3.565×10^{-3}	2.728×10^{-3}	1.837×10^{-3}

$$f_i^* = a_i^n \xi_i^4 + b_i^n \xi_i^3 + c_i^n \xi_i^2 + f_i^n \xi_i + f_i^n,$$

$$f_i'^* = 4a_i^n \xi_i^3 + 3b_i^n \xi_i^2 + 2c_i^n \xi_i + f_i'^n,$$

and, for CIP-CSL2,

$$f_i^* = 3A1_i \xi_i^2 + 2A2_i \xi_i + f_i^n.$$

The nonadvection phase is written as

$$f_i^{n+1} = f_i^* + G\Delta t,$$

$$f_i'^{n+1} = f_i'^* - (u_i'^* f_i'^*)\Delta t + \frac{f_{i+1}^{n+1} - f_{i+1}^* - f_{i-1}^{n+1} + f_{i-1}^*}{2\Delta x}$$

(CIP-CSL4 only).

For the calculation of ρ ($\equiv \int f dx$), we use the conservative form of Eq. (34). Therefore it is solved as

$$\rho_{i-1/2}^* = \Delta \rho_{i-1}^n + (\rho_{i-1/2}^n - \Delta \rho_i^n), \quad (38)$$

$$\rho_{i-1/2}^{n+1} = \rho_{i-1/2}^* + \int_{x_{i-1}}^{x_i} \int_{t^n}^{t^{n+1}} S(x, t) dx dt. \quad (39)$$

In the calculation of Eq. (38), Eqs. (16) and (28) are used.

It is important to note that the mass conservation is recovered in a form of spatial profile within a grid cell every time step when the polynomial is reconstructed under the constraint of Eq. (9), even if mass flux is separately treated in the nonadvection term.

d. Numerical tests

We shall present some sample calculations to test the procedure given in the previous sections. A convergence study was conducted for the schemes presented. A square wave advected with a constant velocity field, the same as that in Fig. 2, was computed with different spatial resolutions. The grid-refinement experiments were conducted by reducing the computational grid spacing as shown in Table 1. The numerical results after $t = 100$ were used for the error estimation. A measure of the total error is calculated by

$$E_{\text{TOT}} = \frac{1}{N} \sum_{i=1}^N (f_i^n - f_i^{\text{exact}})^2, \quad (40)$$

where N is the total number of grid points. It is found that the all the schemes converge to the exact solution when the grid is refined. The original CIP and the CIP-CSL2 give very similar numerical errors as seen visually in Fig. 2. The CIP-CSL4 scheme gives a less diffusive solution and has a higher accuracy. We should note that the present example uses a square wave, and therefore

overshooting error at the edge does not decrease linearly in proportion to mesh size.

The second example is advection with variable velocity and Eq. (8) is solved under a given velocity field,

$$u = 1 + 0.5 \sin(2\pi x/100), \quad (41)$$

with the following initial condition,

$$f(0, x) = \begin{cases} 1 & \text{if } 40 \leq x \leq 60 \\ 0 & \text{otherwise,} \end{cases}$$

where equally spaced grid points of $\Delta x = 100/(N - 1)$ and time step size of $\Delta t = 10/(N - 1)$ are used, N being the number of grid points.

We repeated the calculation by changing the total grid points to $N = 101, 301$, and 1001 to test grid dependence of the scheme proposed here and the results are shown in Fig. 6. All profiles are those after $10(N - 1)$ time steps that corresponds to $t = 100$.

We found that the accuracy has been improved at the discontinuity by the CIP-CSL4 scheme, because it uses a fourth-order polynomial and the CIP-CSL2 gives quite a similar result to the CIP. It is important to note that the first-order upwind scheme needs 10 001 grid points to obtain the same result as the present schemes or the CIP of 101 grid points, which already converged to the solution regardless of grid size. Furthermore, it has already been shown that most of the modern schemes like TVD (Sweby 1984) fail to reproduce the result with the comparable accuracy on 101 grid points (Tanaka et al. 1999).

The following one-dimensional Burgers' equation is an interesting example of application to nonlinear equations:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \lambda \frac{\partial^2 u}{\partial x^2} \equiv S. \quad (42)$$

Since the CIP and CIP-CSL4 need to calculate the spatial derivative of u , we differentiate Eq. (42) as

$$\frac{\partial u'}{\partial t} + u \frac{\partial u'}{\partial x} = -u'^2 + S'. \quad (43)$$

The generalized procedure in the previous section is then applied to these equations.

For the calculation of ρ ($\equiv \int u dx$), we use the remapping procedure given in section 3a combined with the diffusion process:

$$\rho_{i-1/2}^* = \Delta \rho_{i-1}^n + (\rho_{i-1/2}^n - \Delta \rho_i^n), \quad (44)$$

$$\rho_{i-1/2}^{n+1} = \rho_{i-1/2}^* + \frac{\lambda \Delta t}{\Delta x^2} (\rho_{i+1/2}^* - 2\rho_{i-1/2}^* + \rho_{i-3/2}^*). \quad (45)$$

We should note that in the calculation of Eq. (44), Eqs. (16) and (28) are used but with $\xi = -u\Delta t/2$ since the transportation velocity now becomes $u/2$ according to the conservation form of Eq. (42):

$$\frac{\partial u}{\partial t} + \frac{\partial(u^2/2)}{\partial x} = S. \quad (46)$$

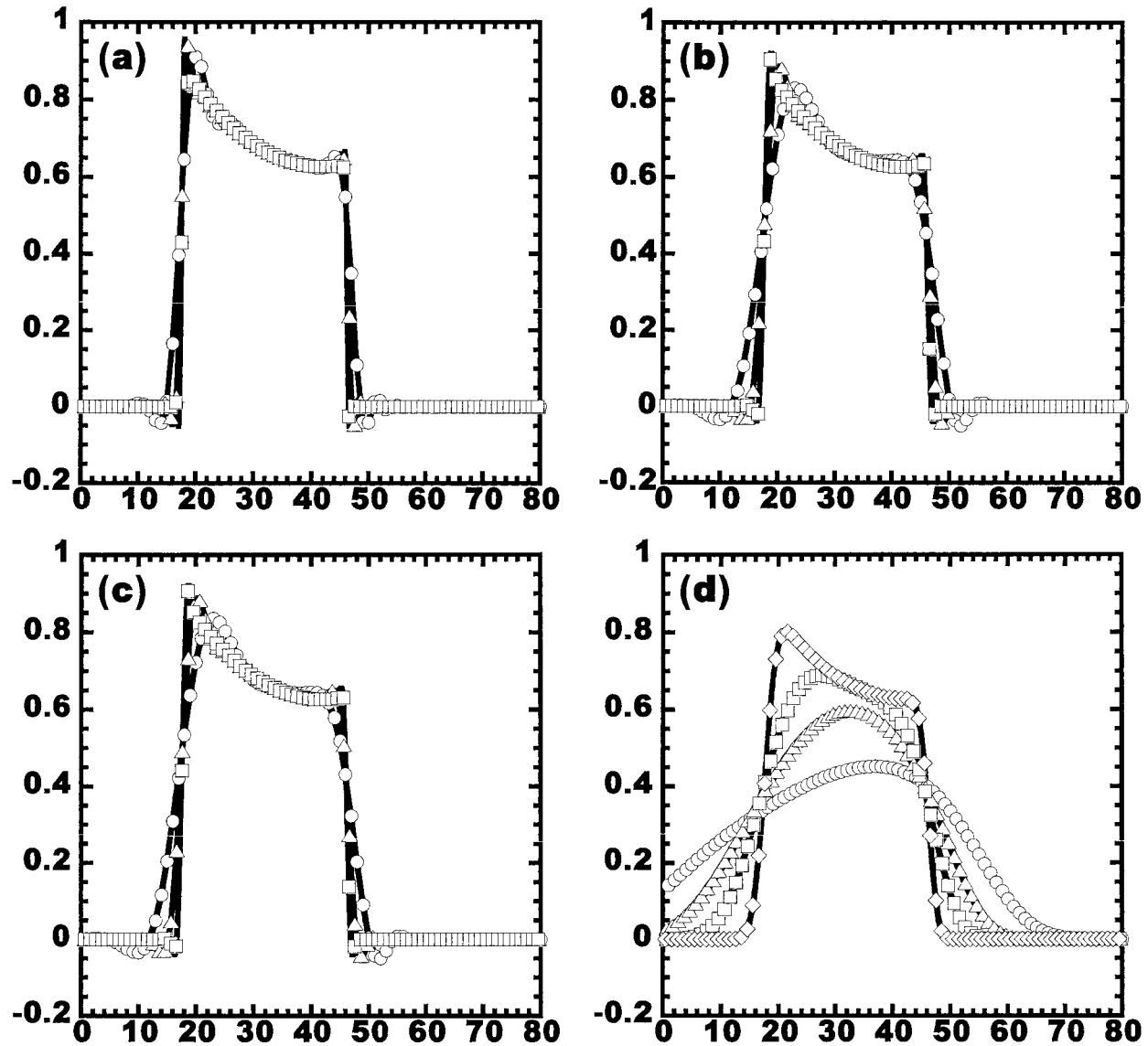


FIG. 6. Propagation of a square wave with a given velocity field by (a) CIP-CSL4, (b) CIP-CSL2, (c) CIP, and (d) the upwind scheme at $t = 100$. The number of grids is 101 (circle), 301 (triangle), 1001 (square), 10 001 (diamond, only for upwind scheme).

The finite-difference approximation for the viscous term is used in Eq. (45), which is also in a conservative form.

Figure 7 shows the calculation result at $t = 100$ with $\lambda = 0$. The initial condition is

$$u(0, x) = 0.5 + 0.4 \cos(2\pi x/100). \quad (47)$$

An equally spaced grid with $\Delta x = 1.0$ and a time step of $\Delta t = 0.1$ are used. The mesh number is $N = 101$.

In order to check the exact speed of a shock wave, we show the result from the first-order upwind scheme with $N = 1001$ as well.

In this calculation, although the viscosity term is not included, the speed of a shock wave is exactly reproduced by the present schemes. The original CIP, how-

ever, was not able to produce a correct shock position without an artificial viscosity because of lacking the conservative feature. Figure 7d shows the results from the original CIP scheme with a proper artificial viscosity. The shock position is correctly reproduced.

4. Calculation with large CFL number

As already stated, the CIP can be used for a large CFL number if the particle trajectory could be successfully traced according to Eq. (6) through variable velocity fields. Various methods have been proposed to find this trajectory (Stainforth and Côté 1991; Smolarkiewicz and Pudykiewicz 1992). We shall use the fourth-order Runge-Kutta method for the time integration of

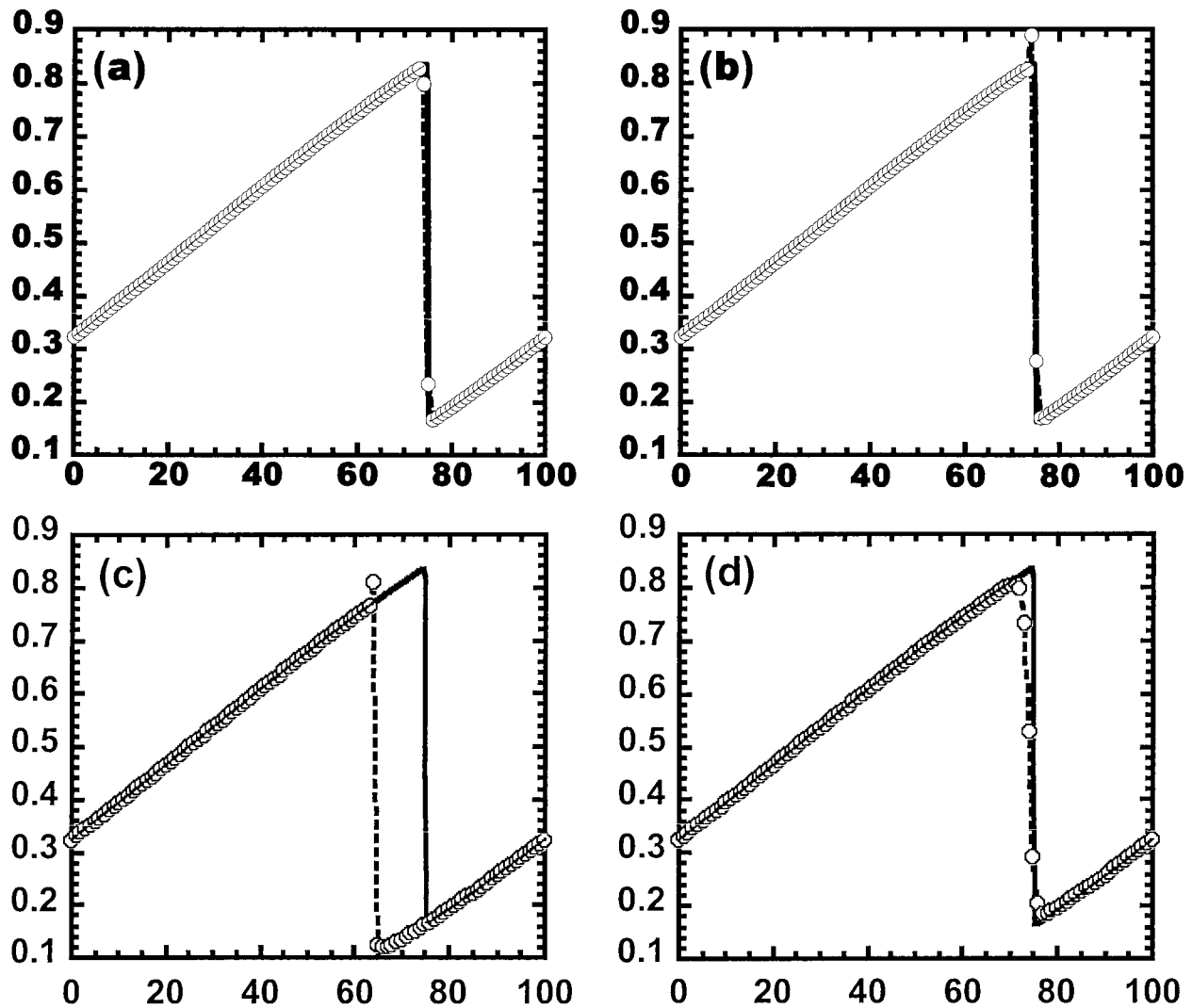


FIG. 7. The results of Burgers's equation without viscosity by (a) CIP-CSL4, (b) CIP-CSL2, and (c) original CIP with 101 grid points at $t = 100$. CIP with an artificial viscosity ($\lambda = 0.2$) is given in (d). For comparison, the result of the first-order upwind scheme with 1001 grid points at $t = 100$ is shown by the solid line.

Eq. (6) although it is usually not necessary in one dimension.

We should note that the density evolution within a cell is also calculated by remapping using the trajectories of four grid points in two dimensions that define a grid cell as already stated in section 3a. This procedure is completely different from the flux-form semi-Lagrangian scheme proposed by Lin and Rood (1996), since in the latter method the flux computed in this way is only first order and is heavily dependent on the smoothness of the velocity field and all the flux through the trajectories must be taken into account.

The target equation (8) contains two different parts: one is advection that can be treated by the above-mentioned semi-Lagrangian procedure, and the other is the compression part like

$$\frac{\partial f}{\partial t} = -f \frac{\partial u}{\partial x}. \quad (48)$$

This part reflects the correction to the mass due to the divergence or convergence of the velocity field. As in any other semi-Lagrangian method, this velocity correction part can be treated as a source term and computed by

$$f_i^{n+1} = f_i^* - \int_T \left(f \frac{\partial u}{\partial x} \right) dt, \quad (49)$$

where f_i^* is the provisional value immediately after the advection phase and T the parcel trajectory connecting the departure and arrival points. Practically, this can be easily done by estimating $f(\partial u/\partial x)$ on every computa-

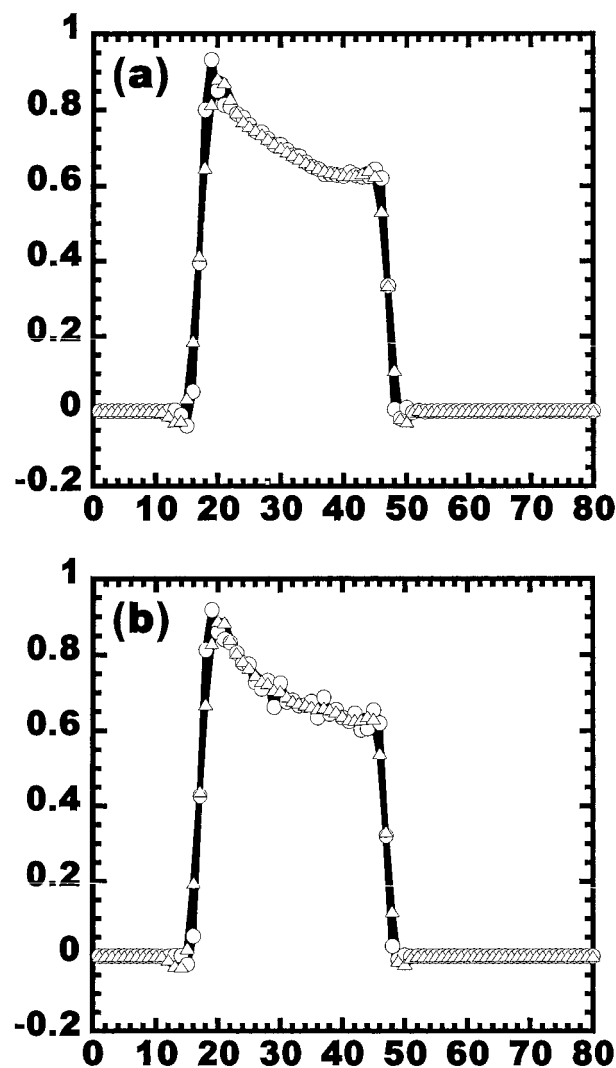


FIG. 8. The same simulation as Fig. 6 but with a large CFL number: (a) CFL = 2 and (b) CFL = 5, by CIP-CSL4 (circle) and CIP-CSL2 (triangle).

tional cell the trajectory passes through and summing them up.

We apply the above procedure to the problem of the variable velocity field specified by Eq. (41). One hundred grid points were used and $\Delta x = 1$. Figure 8 shows some results containing various CFL numbers. Even up to CFL = 5, the scheme works well.

5. Conclusions

We have proposed new algorithms that make nonconservative semi-Lagrangian schemes be exactly conservative. In the mass conservation law, mass is used as an additional variable and it corrects the conservation error originating from a nonconservative formulation by continuously modifying the shape of the interpolation function. By this method, the total mass, which is de-

fined to be the integrated value of f over the space, is exactly conserved.

This scheme has been tested by linear advection with a variable velocity field and Burgers's equation. Although further investigation would improve the result, we believe that being a conservative semi-Lagrangian method, the presented schemes have a great potential for applicability to various atmospheric problems where conservation affects essentially the physics. The present scheme can also provide a useful tool for solving a nonlinear equation in a nonconservative representation that has been proven to be quite stable for multiphase flow calculations.

From the numerical experiment results, we find that the CIP-CSL4 scheme gives a less diffused solution to the discontinuous jump compared with the CSL2 scheme. The jump from the CSL4 scheme covers less than five grid points even after a long time integration. However, the CSL4 scheme requires a more complicated numerical procedure as has been discussed in section 3. Without the calculation for the gradient, the CSL2 scheme appears more economical and easier to be extended to multidimensions.

The extension to multiple dimensions may need a little concern about the introduction of the integrated value of f and the definition of polynomial, and will be given in another paper appearing shortly.

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